

RAJALAKSHMI ENGINEERING COLLEGE

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
CS23334 – FUNDAMENTALS OF DATA SCIENCE LAB (REGULATION 2023)

Experiment No.	18
Experiment Name	Handle Missing Values and Inappropriate Data

DESCRIPTION

To perform the experiment: Handle Missing Values and Inappropriate Data.

CODE

```
import pandas as pd
import numpy as np

df = pd.DataFrame({
    "Age": [22, 35, np.nan, 29],
    "Rating": [4, 6, -1, np.nan],
    "Hotel": ["Ibis", "LemonTree", "Ibys", None]
})
print("Before cleaning:")
print(df)
df["Age"] = df["Age"].fillna(df["Age"].median())
df["Rating"] = df["Rating"].where(df["Rating"].between(1, 5))
valid_median = df.loc[df["Rating"].between(1, 5), "Rating"].median()
df["Rating"] = df["Rating"].fillna(valid_median)
df["Hotel"] = df["Hotel"].replace({"Ibys": "Ibis"})
df["Hotel"] = df["Hotel"].fillna(df["Hotel"].mode()[0])
print("\nAfter cleaning:")
print(df)
print("\nMissing values:")
print(df.isnull().sum())
```

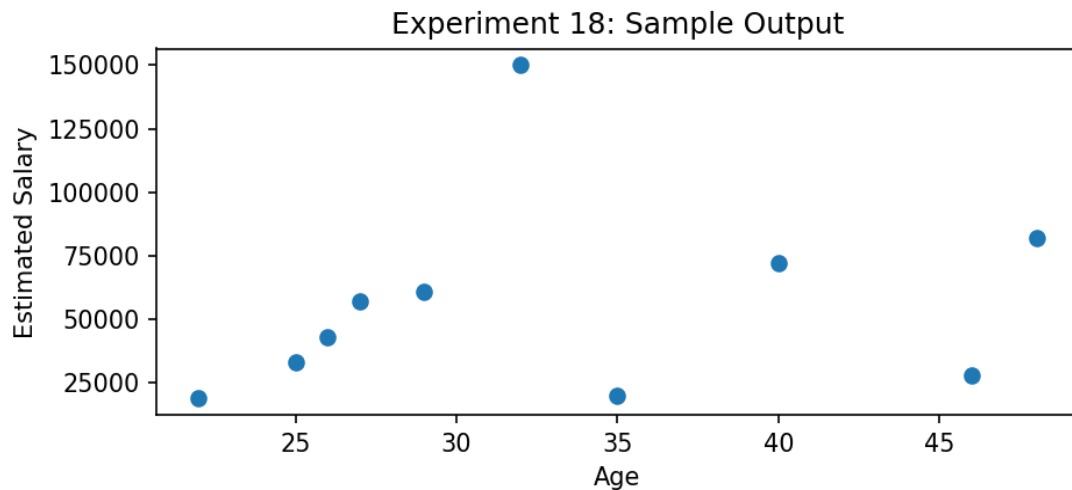
SAMPLE INPUT

Sample input: age and rating have missing values; ratings 6 and -1 are invalid; hotel name "Ibys" is misspelled.

OUTPUT — SAMPLE CONSOLE OUTPUT

Cleaned table printed.\nMissing-value counts after cleaning: 0

OUTPUT DIAGRAM / GRAPH



RESULT

The program demonstrates the stated experiment using the sample data. Verify the output by running the code in your Python environment before academic submission.

Student Signature

Faculty Signature